Hrithik Rai Saxena

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Work Experience - 4+ years

Livello Technologies Düsseldorf, Germany
Data Scientist May 2024 – Aug 2024

- Led development of a revenue and product demand forecasting system incorporating AI-research, backend, and infra.
- Developed CV on edge: Backend and database migration pipelines for a demographic analysis system on jetson Nano.
- Managed production releases on GCP- Cloud Run instances using CI/CD pipelines (Gitlab).
- Reduced Cumulative Forecast Error from 8 to 3.2 by implementing probabilistic modelling for intermittent demand.

Livello Technologies Düsseldorf, Germany
Research Student – Master Thesis Nov 2023 – May 2024

- Spearheaded the development and deployment of dynamically fine-tuned SARIMA-X models across multiple kiosks using multivariate TSA and hyperparameters optimization in Python and R using subprocess-driven data transfer.
- Led the architectural refactoring of forecast microservice for performance and cost optimization.
- Resolved production issues: high response time for huge payloads, memory leaks, system overloads, and downtimes.
- Collaborated with cross-functional teams (QA, Frontend) and assisted in A/B and user acceptance testing (UAT).
- Key achievement – Top Star Performer

Livello Technologies Düsseldorf, Germany
Data science intern Aug 2023 – Nov 2023

- Introduced exogenous variables (holidays, weather) which reduced the average MAPE from 32% to 15%.
- Developed new features for the forecast service – confidence bounds, dashboard business metrics, ranking demand tables.
- Developed ETL pipelines(mongoDB to BigQuery), ensuring data integrity and accessibility for ML tasks.
- Performed code refactoring, optimization and prepared technical documentation on Confluence.

SYSKRON GmbH (Krones) Regensburg, Germany
AI Intern Oct 2022 - Mar 2023

- Formulated model retraining strategies based on data and concept drifts for Reinforcement Learning Product - ContiloopAI.
- Updated data migration and preprocessing pipelines: Implemented feature engineering on raw factory data coming from edge(ReadyKit) via Share2Act to extract unique set points. This saved on-site manual data acquisition field trips.
- Documented training articles on reinforcement learning, model retraining and monitoring for the AI team.

SCHMP, Government of India (MP) Bhopal, India
Data Scientist and Analyst (Contract) Sept 2018 – Feb 2020

- Developed end-to-end machine learning prototypes (SVM classifier) and scaled them to run in a production environment.
- Analyzed old information architectures and contributed to the design and development of new ones.

Education

Deggendorf Institute of Technology Master of Science (M.S.) - Major in Artificial Intelligence and Data Science
Deggendorf, Germany Sept 2021 – Mar 2024

University Institute of Technology, RGPV, India Bachelor of Engineering (B.E.) - Information Technology
Bhopal, India Apr 2015 – May 2019

PROJECTS

- FraudHawk - Real-time Transaction Monitoring and Fraud detection via Autoencoders – Pytorch, Sentence Transformers
- MediSense-AI - Provides personalized treatment plan and cost based on your symptoms – BERT, Pandas,numpy,Pinecone
- Satellite Image Segmentation – Precise land cover classification via U-Net CNN (Keras) and web app (Gradio)
- Langchain projects – RAG based chatbots,LORA fine tuning,Agents,reACT,Chroma,visualizing embeddings via PCA + t-sne

Technical Skills

Programming Languages : Python, R, C

AI: Machine Learning (scikit-learn,statsmodels,scipy,opencv),Deep Learning (PyTorch,TensorFlow,Keras,HuggingFace-Transformers), **GenerativeAI** (LLMs,LORA Fine Tuning,Langchain,RAG,Agents,prompt engineering,TexttoImage)

Databases : SQL,mongoDB,Pinecone,Chroma,Redis ; **Cloud** – AWS (S3,Sagemaker,lambda,Bedrock,airflow,opensearch), GCP

Development Practices & Tools: Object-Oriented Programming, modular SDLC, DSLC, Project management for AI, Agile Development, TDD, CI/CD, JIRA, Docker, MLFlow, Weights & Biases, Ray, Flyte, BentoML, Terraform